Primary Functions:

1. Provide a level surface with sufficient strength to support the imposed loads of people and furniture plus the dead loads of flooring and ceiling.
2. Reduce heat loss from lower floor as required.
3. Provide required degree of sound insulation.
4. Provide required degree of fire resistance.

Basic Construction – a timber suspended upper floor consists of a series of beams or joists supported by load bearing walls sized and spaced to carry all the dead and imposed loads.

Joist Sizing – three methods can be used:

1. Building Regs. Approved Document A – Structure. Refs.
   - BS EN 1991-1-1: Actions on structures. General actions. Densities, self weight, imposed loads for buildings (min. 1·5 kN/m^2 distributed, 1.4 kN concentrated).
   - TRADA publication – Span Tables for Solid Timber Members in Dwellings.

2. Calculation formula:

$$BM = \frac{fbd^2}{6}$$

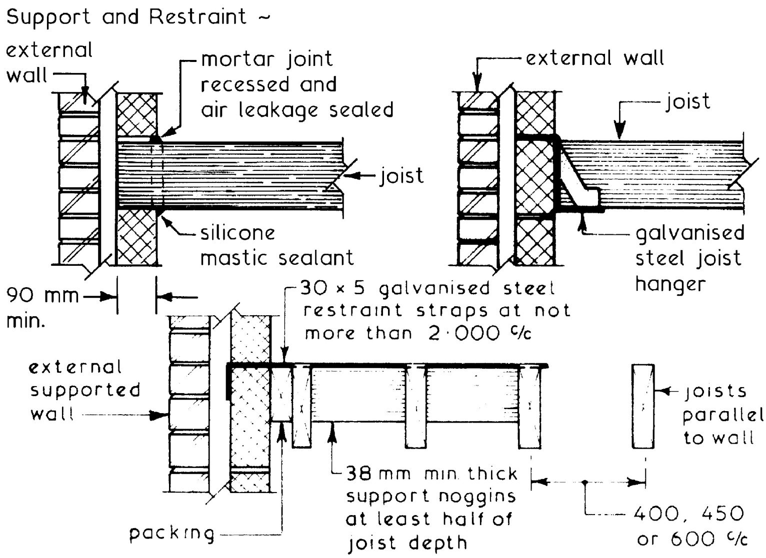
where
BM = bending moment
f = fibre stress
b = breadth
d = depth in mm must be assumed

3. Empirical formula:

$$D = \frac{\text{span in mm}}{24} + 50$$

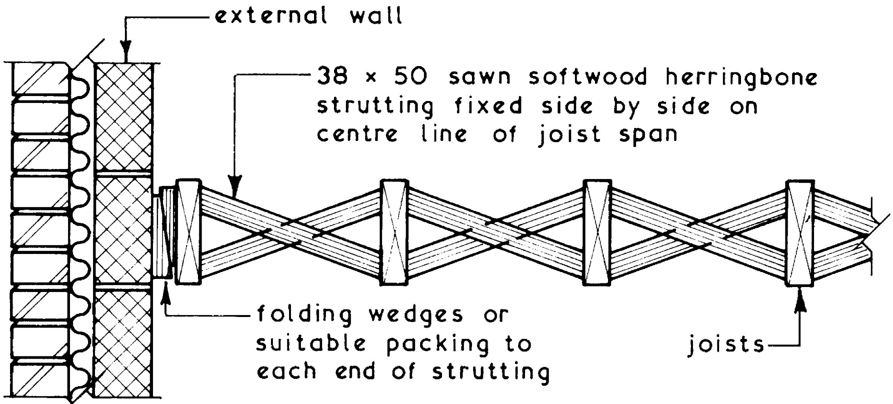
where
D = depth of joist in mm

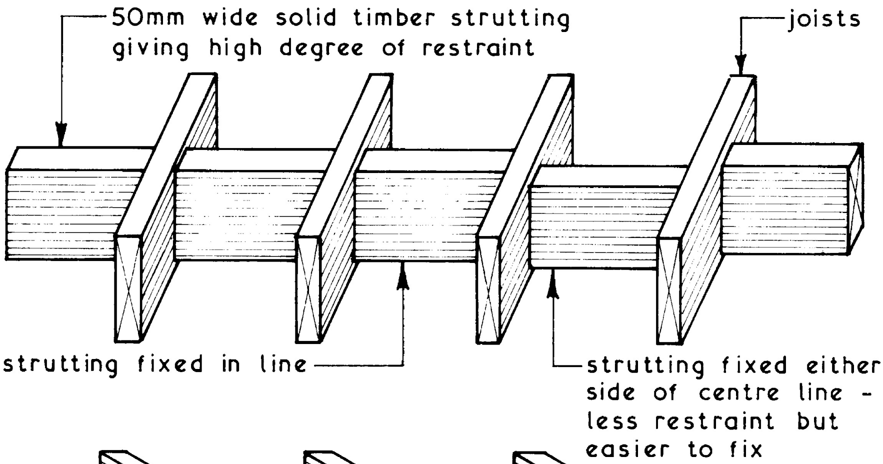
above assumes that joists have a breadth of 50 mm and are at 400 c/c spacing

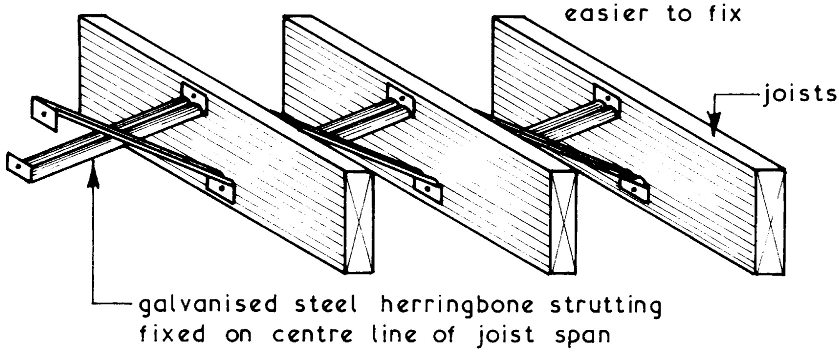


Strutting ~ used in timber suspended floors to restrict the movements due to twisting and vibration which could damage ceiling finishes. Strutting should be included if the span of the floor joists exceeds 2·5m and is positioned on the centre line of the span. Max. floor span ~ 6m measured centre to centre of bearing (inner leaf centre line in cavity wall).

Typical Details ~



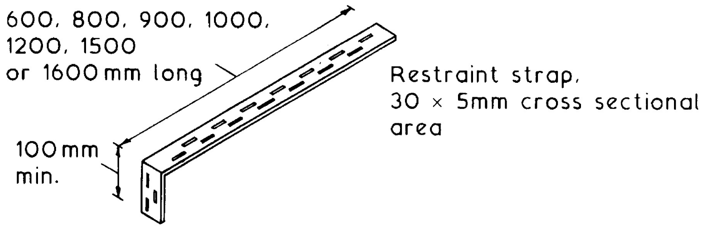




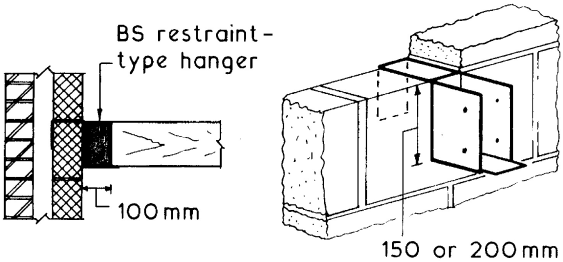
Lateral Restraint ~ external, compartment (fire), separating (party) and internal load bearing walls must have horizontal support from adjacent floors, to restrict movement. Exceptions occur when the wall is less than 3 m long.

Methods:

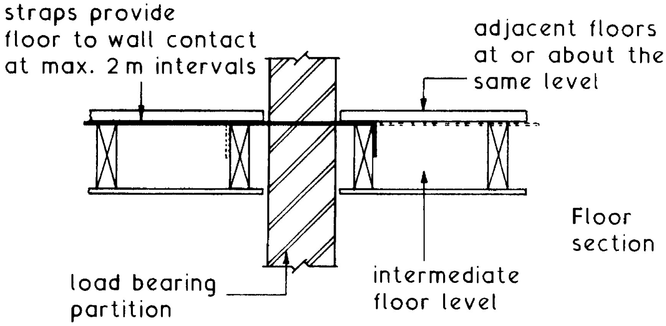
1. 90 mm end bearing of floor joists, spaced not more than 1·2 m apart – see page 813.
2. Galvanised steel straps spaced at intervals not exceeding 2 m and fixed square to joists – see page 813.



3. Joists carried by BS approved galvanised steel hangers.



4. Adjacent floors at or about the same level, contacting with the wall at no more than 2 m intervals.



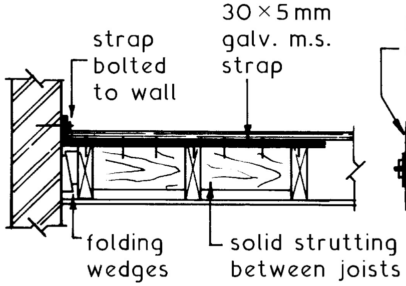
Ref. BS EN 845-1: Specification for ancillary components for masonry. Wall ties, tension straps, hangers and brackets.

Wall Stability – at right angles to floor and ceiling joists this is achieved by building the joists into masonry support walls or locating them on approved joist hangers.

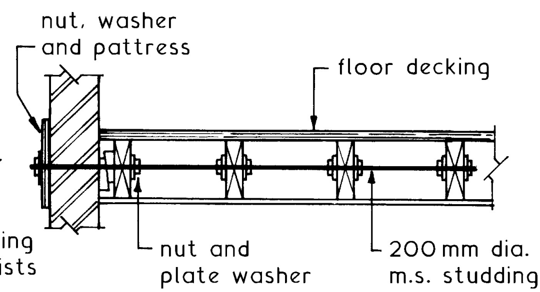
Walls parallel to joists are stabilised by lateral restraint straps. Buildings constructed before current stability requirements (see Bldg. Regs. A.D; A – Structure) often show signs of wall bulge due to the effects of eccentric loading and years of thermal movement.

Remedial Measures –

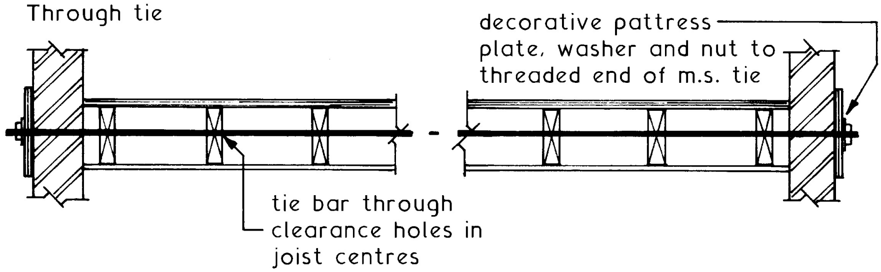
Retro-strap

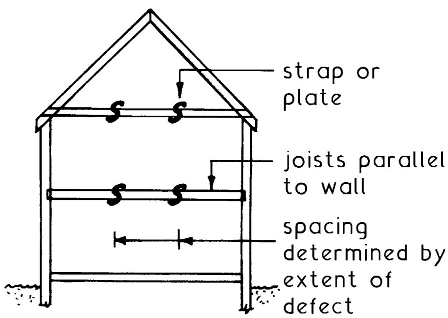


Retro-stud

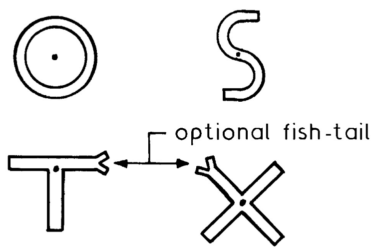


Through tie





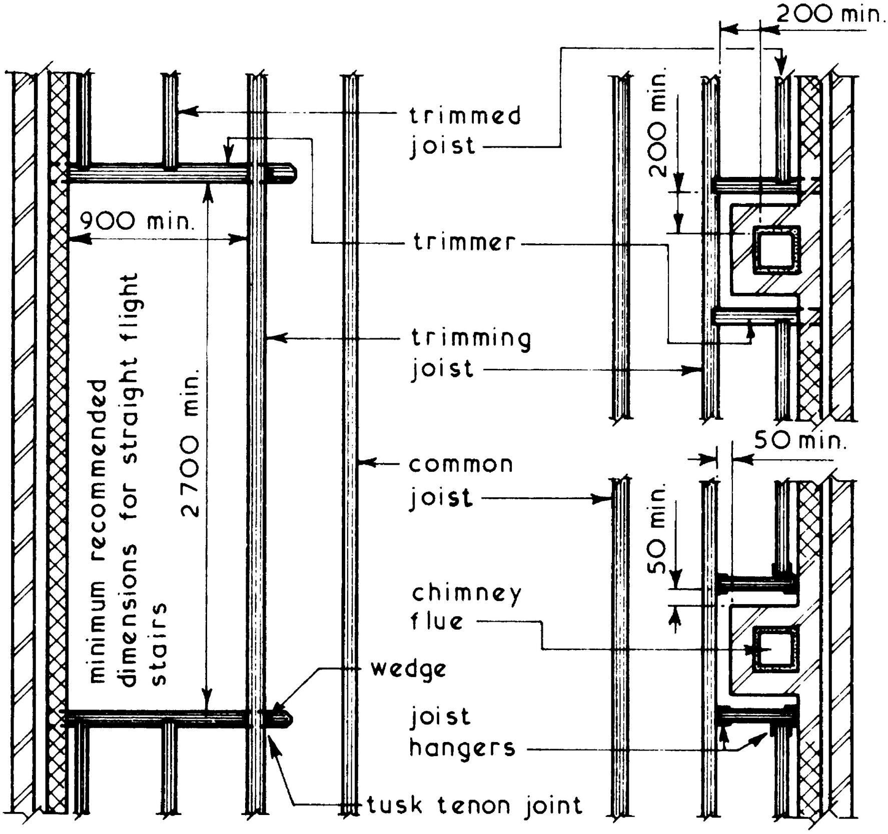
Location of straps and ties



Common decorative iron pattress plates and straps

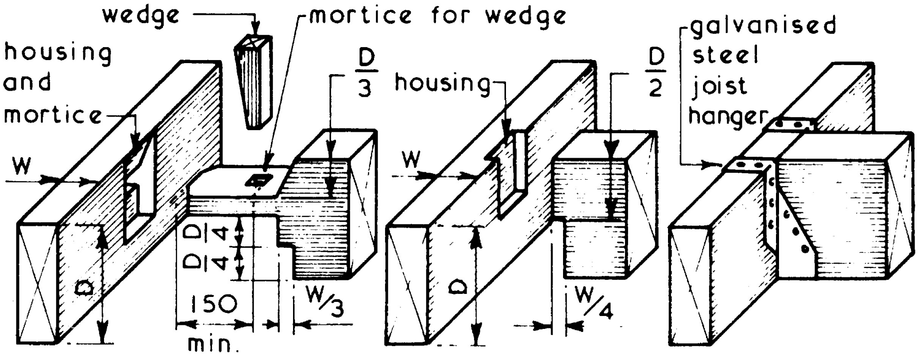
Trimming Members ~ these are the edge members of an opening in a floor and are the same depth as common joists but are usually 25 mm wider.

Typical Details ~



TRIMMING TO STAIRWELL

TRIMMING AROUND FLUE



TRADITIONAL TUSK TENON JOINT

TRADITIONAL HOUSED JOINT

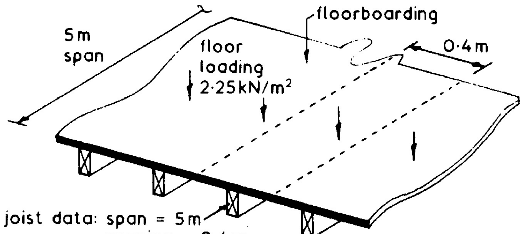
MODERN JOIST HANGER

Typical spans and loading for floor joists of general structural (GS) grade:

	Dead weight of flooring and ceiling, excluding the self-weight of the joists (kg/m^2)								
	< 25			25–50			50–125		
	Spacing of joists (mm c/c)								
	400	450	600	400	450	600	400	450	600
Sawn size (mm × mm)	Maximum clear span (m)								
38 × 75	1.22	1.09	0.83	1.14	1.03	0.79	0.98	0.89	0.70
38 × 100	1.91	1.78	1.38	1.80	1.64	1.28	1.49	1.36	1.09
38 × 125	2.54	2.45	2.01	2.43	2.30	1.83	2.01	1.85	1.50
38 × 150	3.05	2.93	2.56	2.91	2.76	2.40	2.50	2.35	1.93
38 × 175	3.55	3.40	2.96	3.37	3.19	2.77	2.89	2.73	2.36
38 × 200	4.04	3.85	3.35	3.82	3.61	3.13	3.27	3.09	2.68
38 × 225	4.53	4.29	3.73	4.25	4.02	3.50	3.65	3.44	2.99
50 × 75	1.45	1.37	1.08	1.39	1.30	1.01	1.22	1.11	0.88
50 × 100	2.18	2.06	1.76	2.06	1.95	1.62	1.82	1.67	1.35
50 × 125	2.79	2.68	2.44	2.67	2.56	2.28	2.40	2.24	1.84
50 × 150	3.33	3.21	2.92	3.19	3.07	2.75	2.86	2.70	2.33
50 × 175	3.88	3.73	3.38	3.71	3.57	3.17	3.30	3.12	2.71
50 × 200	4.42	4.25	3.82	4.23	4.07	3.58	3.74	3.53	3.07
50 × 225	4.88	4.74	4.26	4.72	4.57	3.99	4.16	3.94	3.42
63 × 100	2.41	2.29	2.01	2.28	2.17	1.90	2.01	1.91	1.60
63 × 125	3.00	2.89	2.63	2.88	2.77	2.52	2.59	2.49	2.16
63 × 150	3.59	3.46	3.15	3.44	3.31	3.01	3.10	2.98	2.63
63 × 175	4.17	4.02	3.66	4.00	3.85	3.51	3.61	3.47	3.03
63 × 200	4.73	4.58	4.18	4.56	4.39	4.00	4.11	3.95	3.43
63 × 225	5.15	5.01	4.68	4.99	4.85	4.46	4.62	4.40	3.83
75 × 125	3.18	3.06	2.79	3.04	2.93	2.67	2.74	2.64	2.40
75 × 150	3.79	3.66	3.33	3.64	3.50	3.19	3.28	3.16	2.86
75 × 175	4.41	4.25	3.88	4.23	4.07	3.71	3.82	3.68	3.30
75 × 200	4.92	4.79	4.42	4.77	4.64	4.23	4.35	4.19	3.74
75 × 225	5.36	5.22	4.88	5.20	5.06	4.72	4.82	4.69	4.16

Notes:
1. Where a bath is supported, the joists should be duplicated.
2. See pages 39 and 40 for material dead weights.
3. See pages 145 and 146 for softwood classification and grades.

Joist and Beam Sizing ~ design tables and formulae have limitations, therefore where loading, span and/or conventional joist spacings are exceeded, calculations are required. BS EN 1995-1-1: Design of timber structures and BS EN 338: Structural timber – strength classes are both useful resource material for detailed information on a variety of timber species. The following example serves to provide guidance on the design process for determining joist size, measurement of deflection, safe bearing and resistance to shear force:

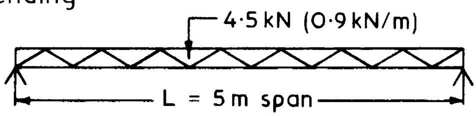


joist data: span = 5 m
spacing = 0·4 m
fibre stress (f) = 7·5 N/mm²
compression perpendicular to grain = 2.4 N/mm²
shear parallel to grain (v) = 0.71 N/mm²
modulus of elasticity (E) = 10,800 N/mm²

Total load (W) per joist = 5 m × 0·4 m × 2·25 kN/m² = 4·5 kN

$$\text{or}: \frac{4{\cdot}5\text{ kN}}{5\text{ m span}} = 0{\cdot}9\text{ kN/m}$$

Resistance to bending ~



Bending moment formulae are shown on pages 678 and 682

$$BM = \frac{WL}{8} = \frac{fbd^2}{6}$$

Where: W = total load, 4·5 kN (4500 N)
L = span, 5 m (5000 mm)
f = fibre stress of timber, 7·5 N/mm²
b = breadth of joist, try 50 mm
d = depth of joist, unknown

Transposing:

$$\frac{WL}{8} = \frac{fbd^2}{6}$$

Becomes:

$$d = \sqrt{\frac{6WL}{8fb}} = \sqrt{\frac{6 \times 4500 \times 5000}{8 \times 7{\cdot}5 \times 50}} = 212\text{ mm}$$

Nearest commercial size: 50 mm × 225 mm

Joist and Beam Sizing ~ calculating overall dimensions alone is insufficient. Checks should also be made to satisfy: resistance to deflection, adequate safe bearing and resistance to shear.

Deflection – should be minimal to prevent damage to plastered ceilings. An allowance of up to 0·003 × span is normally acceptable; for the preceding example this will be:

$0{\cdot}003 \times 5000\,\text{mm} = 15\,\text{mm}$

The formula for calculating deflection due to a uniformly distributed load (see pages 680 and 682) is:

$$\frac{5WL^3}{384EI} \quad \text{where} \quad I = \frac{bd^3}{12}$$

$$I = \frac{50 \times (225)^3}{12} = 4{\cdot}75 \times (10)^7$$

$$\text{So, deflection} = \frac{5 \times 4500 \times (5000)^3}{384 \times 10800 \times 4{\cdot}75 \times (10)^7} = 14{\cdot}27\,\text{mm}$$

NB. This is only just within the calculated allowance of 15 mm, therefore it would be prudent to specify slightly wider or deeper joists to allow for unknown future use.

Safe Bearing ~

$$= \frac{\text{load at the joist end, } W/2}{\text{compression perpendicular to grain} \times \text{breadth}}$$

$$= \frac{4500/2}{2{\cdot}4 \times 50} = 19\,\text{mm}.$$

therefore full support from masonry (90 mm min.) or joist hangers will be more than adequate.

Shear Strength ~

$$V = \frac{2bdv}{3}$$

where: V = vertical loading at the joist end, W/2
v = shear strength parallel to the grain, 0.71 N/mm^2

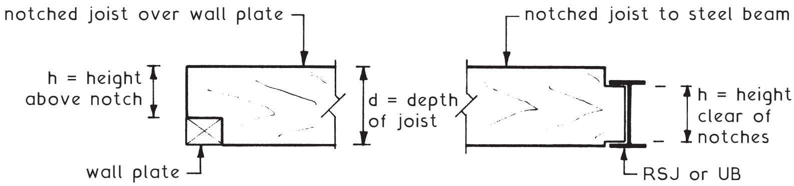
Transposing:-

$$bd = \frac{3V}{2v} = \frac{3 \times 2250}{2 \times 0{\cdot}71} = 4753\,\text{mm}^2 \text{ minimum}$$

Actual bd = 50 mm × 225 mm = 11,250 mm^2

Resistance to shear is satisfied as actual is well above the minimum.

Typical Situations ~



It is necessary to ensure enough timber above and/or below a notch to resist horizontal shear or shear parallel to the grain.
To check whether a joist or beam has adequate horizontal shear strength:

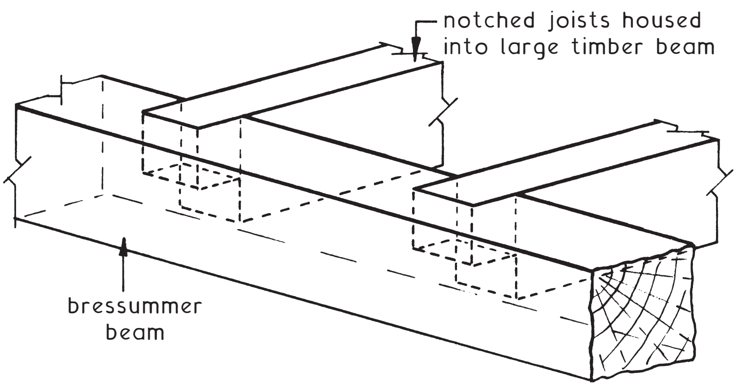
$$v = (V3d) \div (2bh^2)$$

Using the data provided in the previous two pages as applied to the design of a timber joist of 225 × 50 mm cross section, in this instance with a 50 mm notch to leave 175 mm (h) clear:

$$v = (2250 \times 3 \times 225) \div (2 \times 50 \times 175 \times 175) = 0{\cdot}496\,N/mm^2$$

Shear strength parallel to the grain* is given as 0.710 N/mm^2 so sufficient strength is still provided.
*see page 146.

Bressummer (or breastsummer) Beam ~

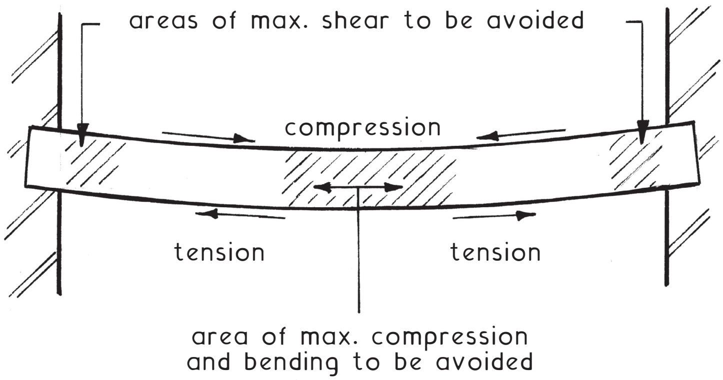


A long, large section timber beam carrying joists directly or notched and housed (as shown). A dated form of construction for supporting a masonry wall and floor/ceiling joists in large openings. Often found during refurbishment work over shop windows. Steel or reinforced concrete now preferred.

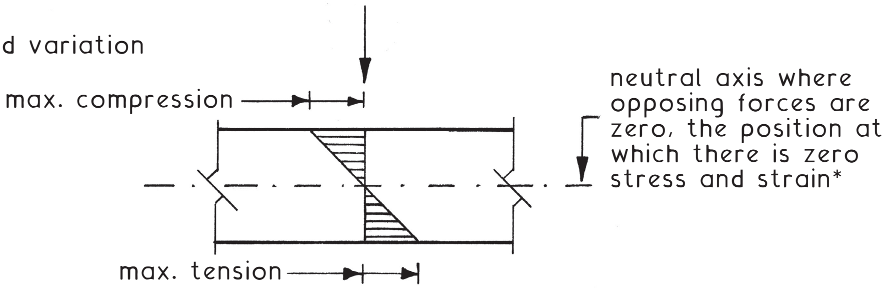
Pipes and cables ~ should be routed discretely and out of sight, but accessible for maintenance and repair. Where timber joists are used, services running parallel can be conveniently attached to the sides. At right angles the timber will need to be holed or notched.

Holing ~ suitable for flexible cables and coiled soft copper micro-bore tubing. The preferred location in simply supported end-bearing floor joists is in the neutral axis. This is where compressive and tensile load distribution neutralises at the centre and where the material gets neither longer nor shorter when deflected.

Joist under load



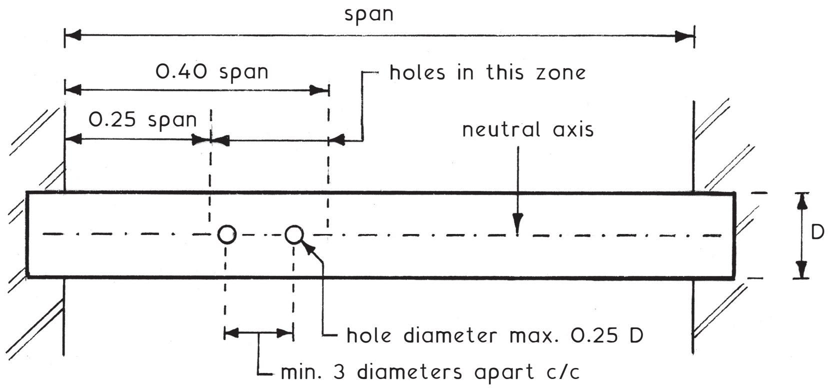
Load variation



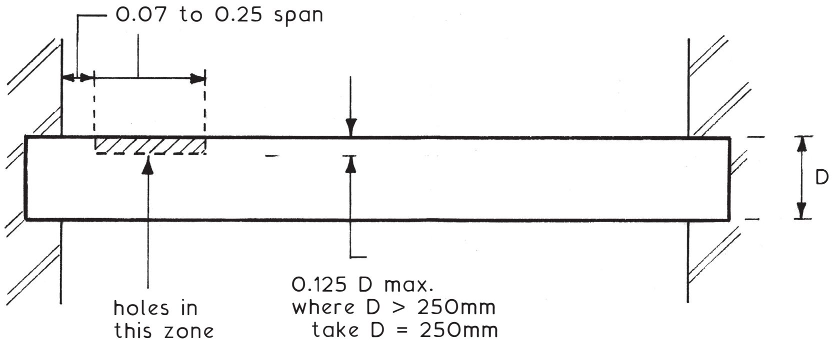
*stress is represented by: loading ÷ cross sectional area
strain is represented by: extension ÷ original length

NB. The relationship between stress and strain is most obvious with metals and plastics. When these are deformed by an applied force they return to their original size and shape after the force is removed. The two physical states of stress and strain are proportional within the elastic state just described. This is known as Hooke's Law and is described by the term elasticity measured in units of N/mm^2. Examples for softwood timber are given on page 146.

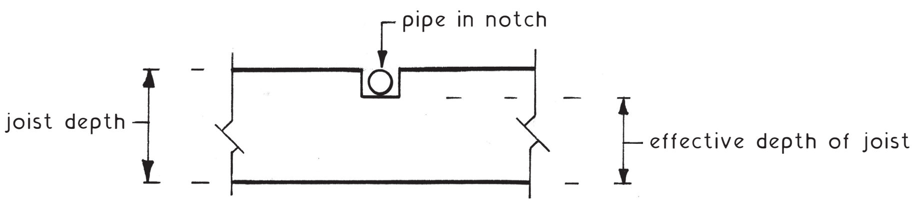
Hole limitations ~
Diameter maximum: 0.25 x joist depth.
Spacing minimum: 3 x diameter apart, measured centre to centre.
Position in the neutral axis: between 0.25 and 0.40 x clear span, measured from the support.



Notching ~ the only practical way of accommodating rigid pipes and conduits in joisted floors.
Depth maximum: 0.125 x joist depth.
Position between: 0.07 and 0.25 x clear span measured from the support.



Notching of a joist reduces the effective depth, thereby weakening the joist and reducing its design strength. To allow for this, joists should be oversized.



For fire protection, floors are categorised depending on their height relative to adjacent ground ~

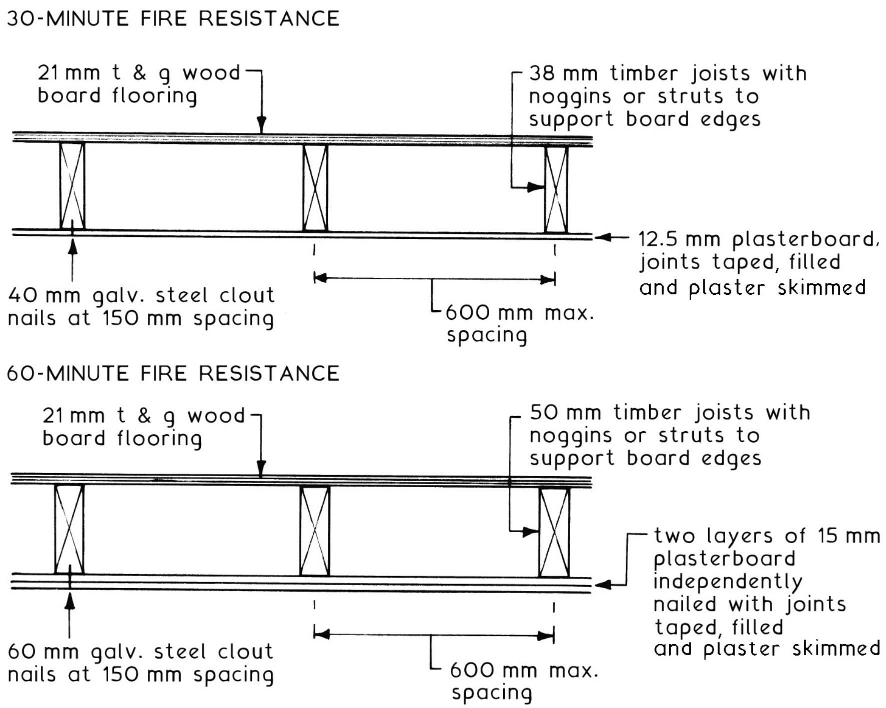
Height of top floor above ground	Fire resistance (load bearing)
Less than 5 m	30 minutes
More than 5 m	60 minutes (30 min. for a three-storey dwelling)

Tests for fire resistance relate to load bearing capacity, integrity and insulation as determined by BS 476 – 21: Fire tests on building materials and structures. Methods for determination of the fire resistance of load bearing elements of construction.

Integrity ~ the ability of an element to resist fire penetration.

Insulation ~ ability to resist heat penetration so that fire is not spread by radiation and conduction.

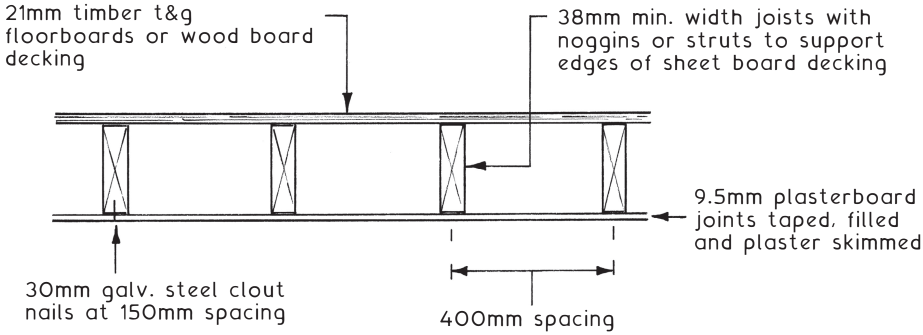
Typical applications ~



Modified 30-minute fire resistance ~ a lower specification than the full 30-minute fire resistance shown on the preceding page. This is acceptable in certain situations such as some loft conversion floors in existing dwellings and in new-build dwellings of limited height. These situations are defined in Section 4.7 and Appendix A, Table A1 of Approved Document B1 to the Building Regulations.

Load bearing resistance to collapse – 30 minutes minimum.
Integrity – 15 minutes minimum.
Insulation – 15 minutes minimum.

MODIFIED 30-MINUTE FIRE RESISTANCE



Established floor construction ~ modified 30 minute fire resistant floor construction is often found during refurbishment of housing constructed during the latter part of the 20th century. Here, 9.5 mm plasterboard lath (rounded edge board 1200 mm long x 406 mm wide – now obsolete) was used with floor joists at 400 mm spacing. It is now the general practice to use a 12.5 mm plasterboard ceiling lining to dwelling house floors as this not only provides better fire resistance, but is more stable.

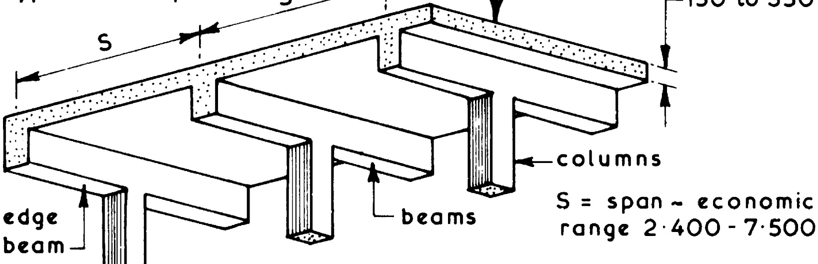
Note 1: Full 30-minute fire resistance must be used for floors over basements or garages.

Note 2: Where a floor provides support or stability to a wall or vice versa, the fire resistance of the supporting element must not be less than the fire resistance of the other element.

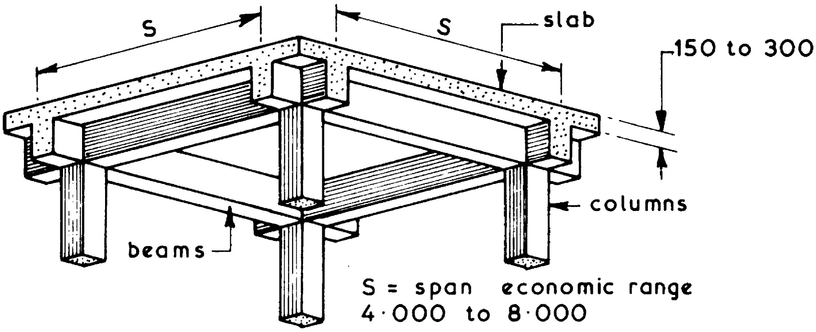
Ref. Building Regulations, AD B Fire safety, Volume 1 – Dwelling houses.

Reinforced Concrete Suspended Floors ~ a simple reinforced concrete flat slab cast to act as a suspended floor is not usually economical for spans over 5·000. To overcome this problem beams can be incorporated into the design to span in one or two directions. Such beams usually span between columns which transfers their loads to the foundations. The disadvantages of introducing beams are the greater overall depth of the floor construction and the increased complexity of the formwork and reinforcement. To reduce the overall depth of the floor construction flat slabs can be used where the beam is incorporated with the depth of the slab. This method usually results in a deeper slab with complex reinforcement especially at the column positions.

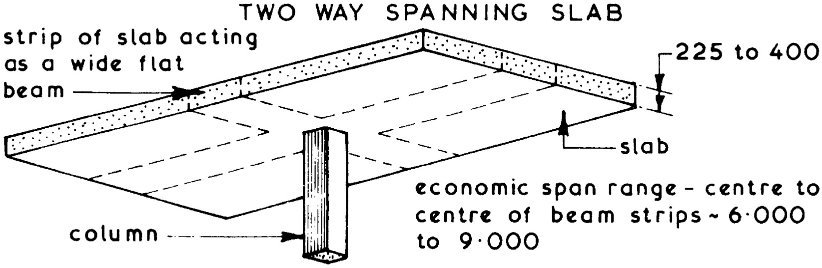
Typical Examples ~



ONE WAY SPANNING SLAB



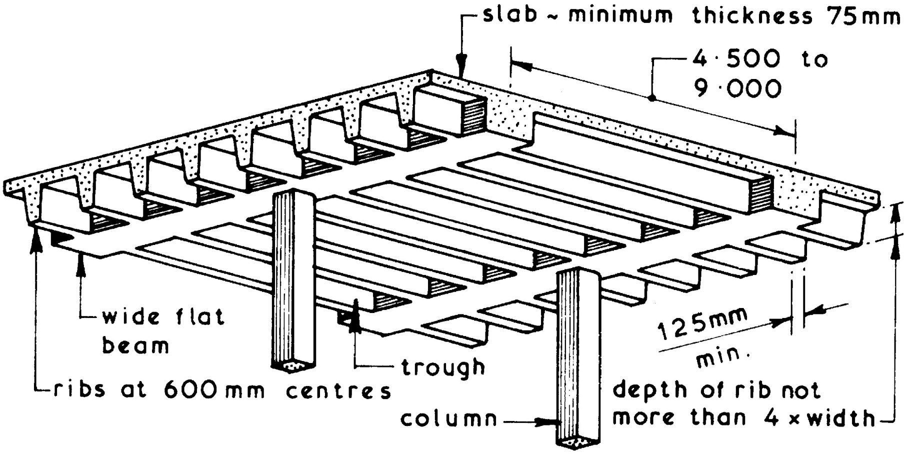
TWO WAY SPANNING SLAB



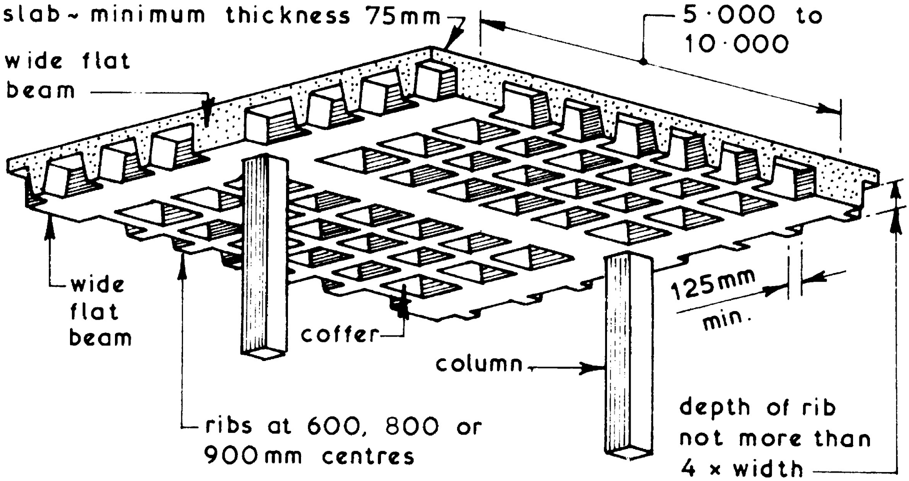
TWO WAY SPANNING FLAT SLAB

Ribbed Floors ~ to reduce the overall depth of a traditional cast in-situ reinforced concrete beam and slab suspended floor a ribbed floor could be used. The basic concept is to replace the wide spaced deep beams with narrow spaced shallow beams or ribs which will carry only a small amount of slab loading. These floors can be designed as one or two way spanning floors. One way spanning ribbed floors are sometimes called troughed floors whereas the two way spanning ribbed floors are called coffered or waffle floors. Ribbed floors are usually cast against metal, glass-fibre or polypropylene preformed moulds which are temporarily supported on plywood decking, joists and props – see page 620.

Typical Examples ~



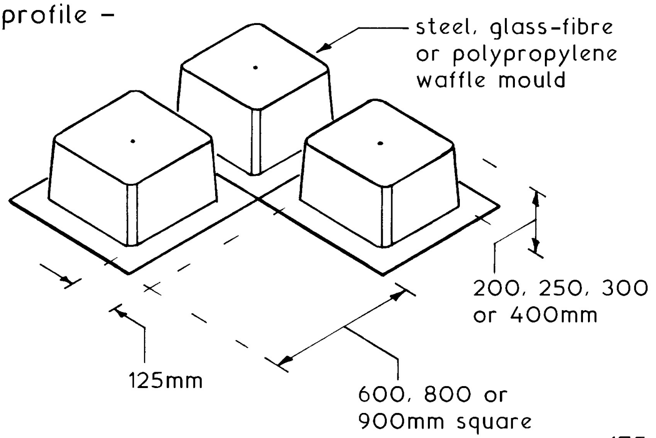
ONE WAY SPANNING RIBBED OR TROUGHED FLOOR

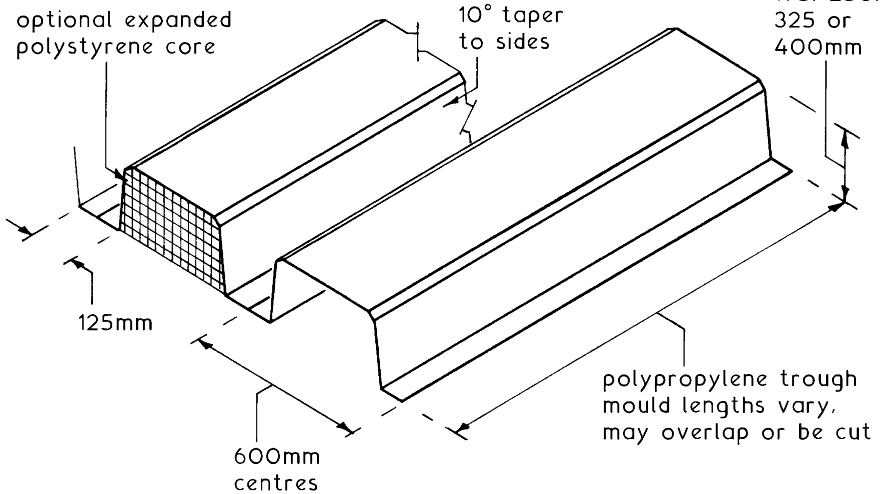


TWO WAY SPANNING COFFERED OR WAFFLE FLOOR

Ribbed Floors – these have greater span and load potential per unit weight than flat slab construction. This benefits a considerable reduction in dead load, to provide cost economies in other super-structural elements and foundations. The regular pattern of voids created with waffle moulds produces a honeycombed effect, which may be left exposed in utility buildings such as car parks. Elsewhere, such as shopping malls, a suspended ceiling would be appropriate. The trough finish is also suitable in various situations and has the advantage of creating a continuous void for accommodation of service cables and pipes. A suspended ceiling can add to this space where air conditioning ducting is required, also providing several options for finishing effect.

Typical mould profile –

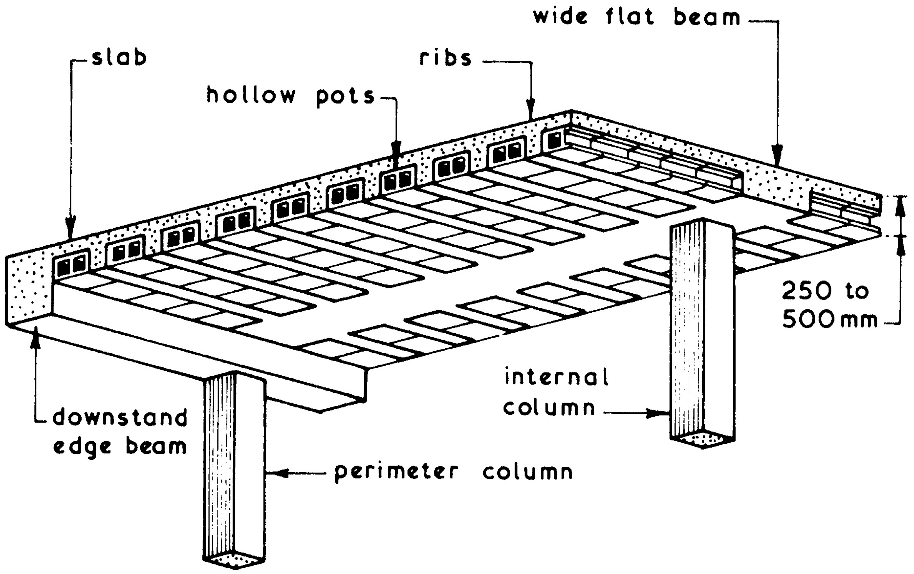




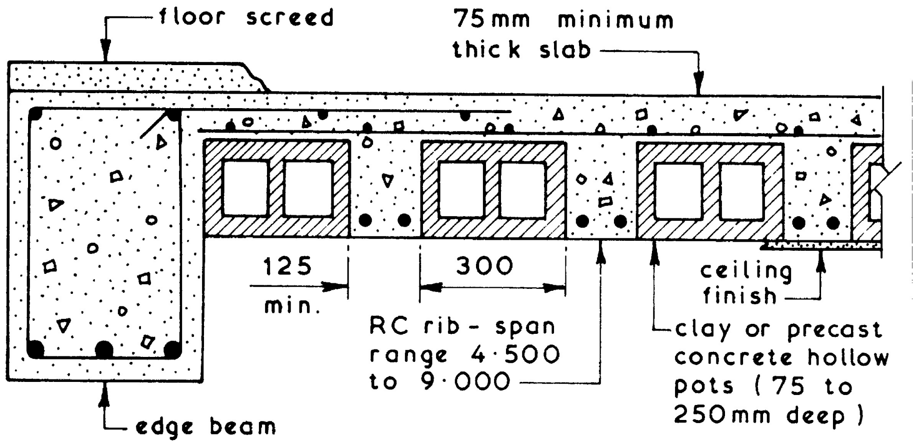
NB. After removing the temporary support structure, moulds are struck by flexing with a flat tool. A compressed air line is also effective.

Hollow Pot Floors ~ these are in essence a ribbed floor with permanent formwork in the form of hollow clay or concrete pots. The main advantage of this type of cast in-situ floor is that it has a flat soffit which is suitable for the direct application of a plaster finish or an attached dry lining. The voids in the pots can be utilised to house small diameter services within the overall depth of the slab. These floors can be designed as one or two way spanning slabs, the common format being the one way spanning floor.

Typical Example ~



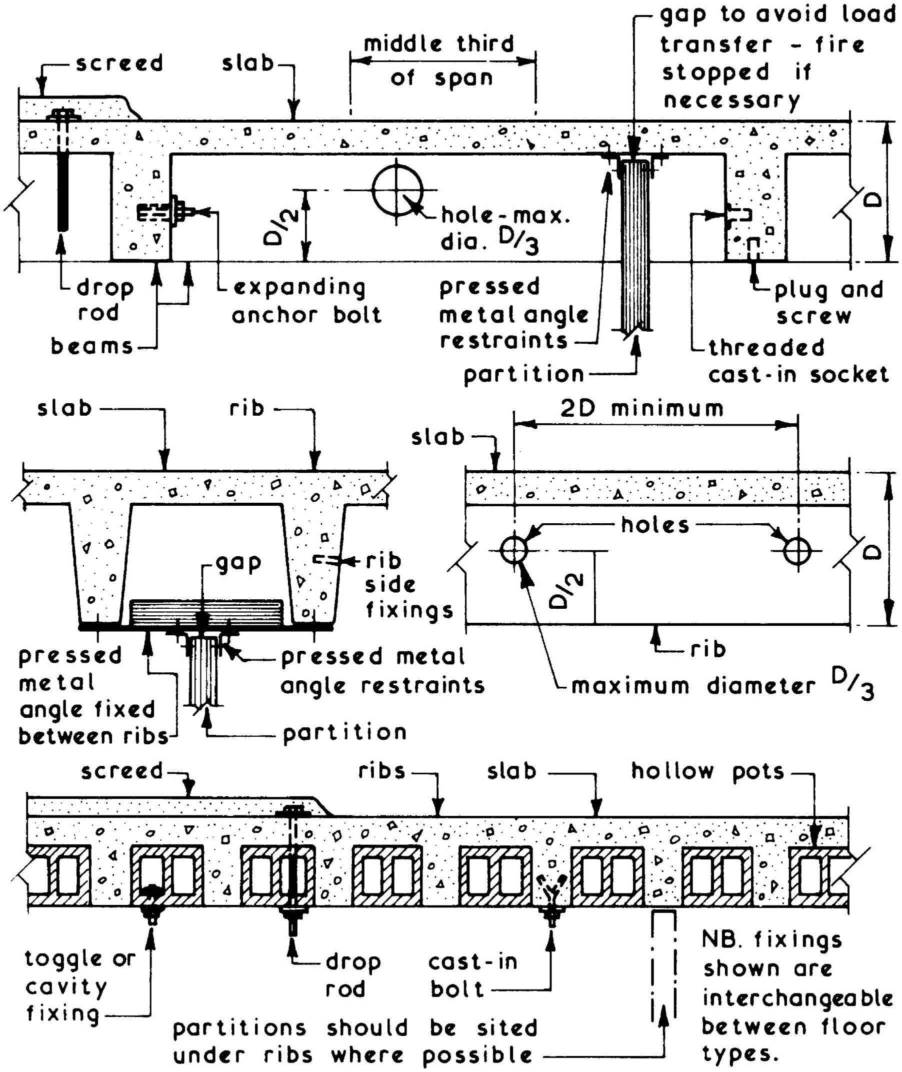
ONE WAY SPANNING HOLLOW POT FLOOR



TYPICAL SECTION

Soffit and Beam Fixings ~ concrete suspended floors can be designed to carry loads other than the direct upper surface loadings. Services can be housed within the voids created by the beams or ribs and suspended or attached ceilings can be supported by the floor. Services which run at right angles to the beams or ribs are usually housed in cast-in holes. There are many types of fixings available for use in conjunction with floor slabs. Some are designed to be cast-in whilst others are fitted after the concrete has cured. All fixings must be positioned and installed so that they are not detrimental to the structural integrity of the floor.

Typical Examples ~



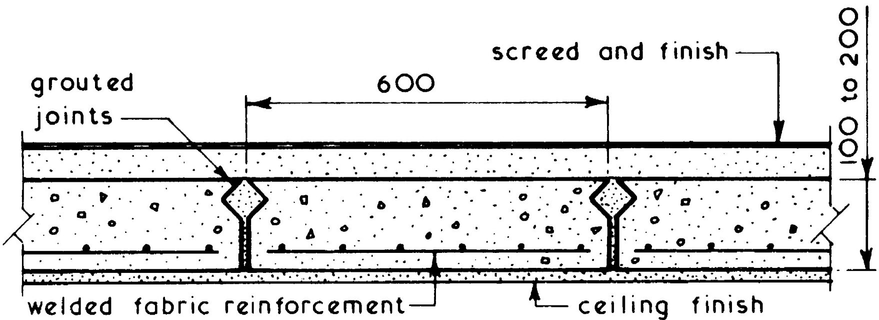
Precast Concrete Floors ~ these are available in several basic formats and provide an alternative form of floor construction to suspended timber floors and in-situ reinforced concrete suspended floors. The main advantages of precast concrete floors are:

1. Elimination of the need for formwork except for nominal propping which is required with some systems.
2. Curing time of concrete is eliminated, therefore the floor is available for use as a working platform at an earlier stage.
3. Superior quality control of product is possible with factory-produced components.

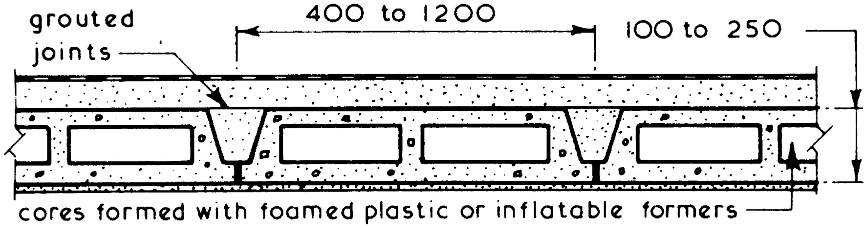
The main disadvantages of precast concrete floors when compared with in-situ reinforced concrete floors are:

1. Less flexible in design terms.
2. Formation of large openings in the floor for ducts, shafts and stairwells usually have to be formed by casting an in-situ reinforced concrete floor strip around the opening position.
3. Higher degree of site accuracy is required to ensure that the precast concrete floor units can be accommodated without any alterations or making good.

Typical Basic Formats ~

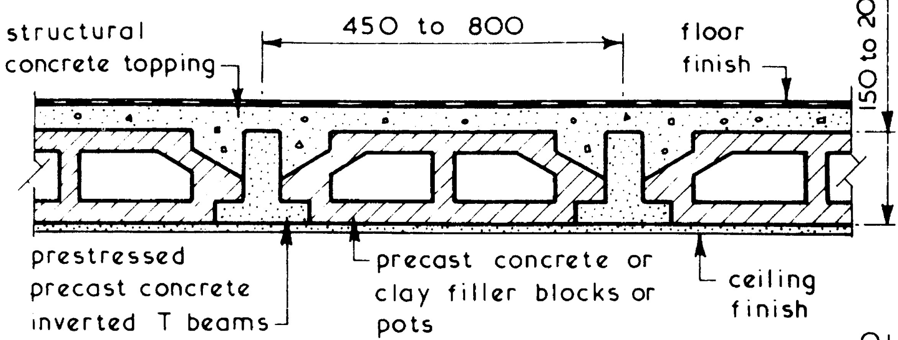


with imposed loading of $1{\cdot}5\,kN/m^2$ spans up to 4·400 max.
SOLID AERATED CONCRETE – usual application roof slabs

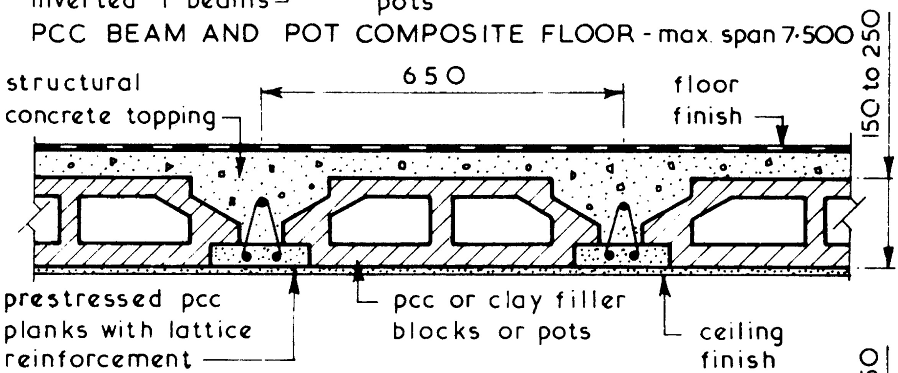


HOLLOW BOX SECTIONS – spans up to 10·000

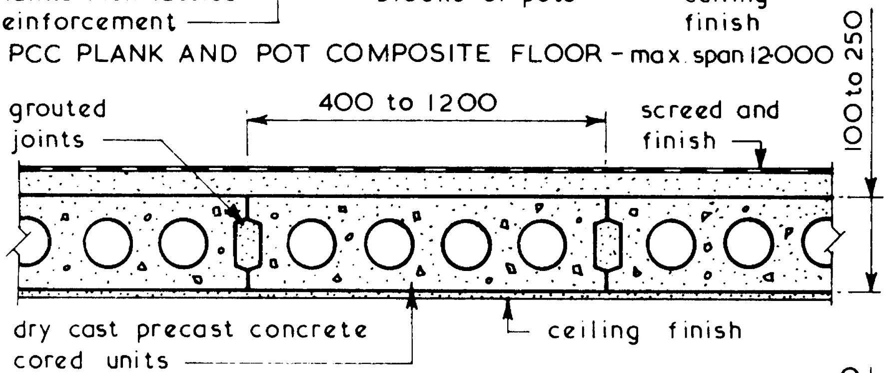
Typical Basic Formats ~



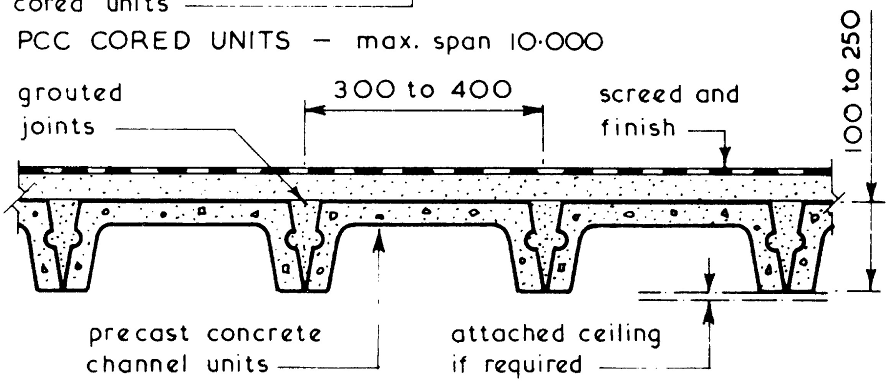
PCC BEAM AND POT COMPOSITE FLOOR - max. span 7·500



PCC PLANK AND POT COMPOSITE FLOOR - max. span 12·000



PCC CORED UNITS — max. span 10·000



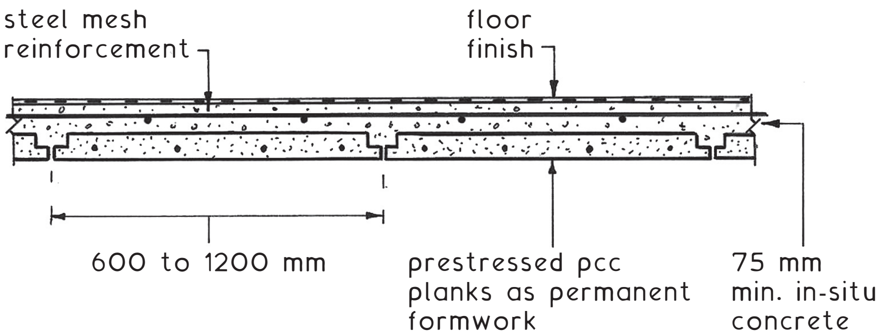
PCC CHANNEL UNITS - max. span 6·750

Composite Floors ~ two types:

1. Narrow prestressed precast concrete beams at about 600 mm spacing; variation occuring with different manufacturers' systems. Concrete filler blocks or "pots" bridge the beams. Blocks and beams are superimposed with an in-situ concrete topping (steel mesh reinforced if required) to complete the structural element. The upper two illustrations on the previous page show examples.

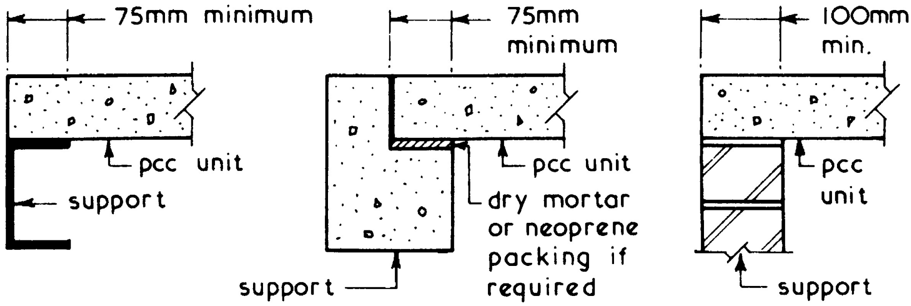
2. Prestressed precast concrete planks produced in widths of 600 to 1200 mm. These are relatively thin and require a structural topping of in-situ steel mesh reinforced concrete. Planks combine with the in-situ concrete as a structural element as well as providing permanent formwork.

During placement of wet concrete both applications will require temporary support from vertical props at a maximum of 2.4 m spacing. Props should remain in place until the concrete has set and cured.



PRESTRESSED PCC PLANKS– spans to 10.000

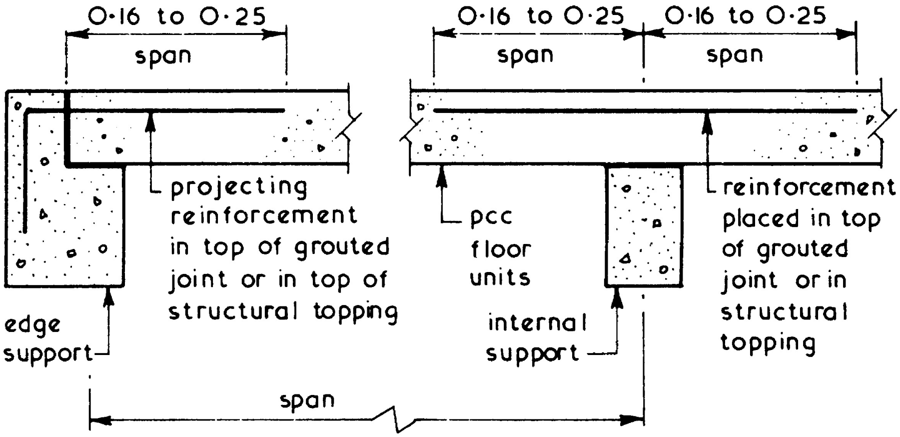
Bearing Considerations ~



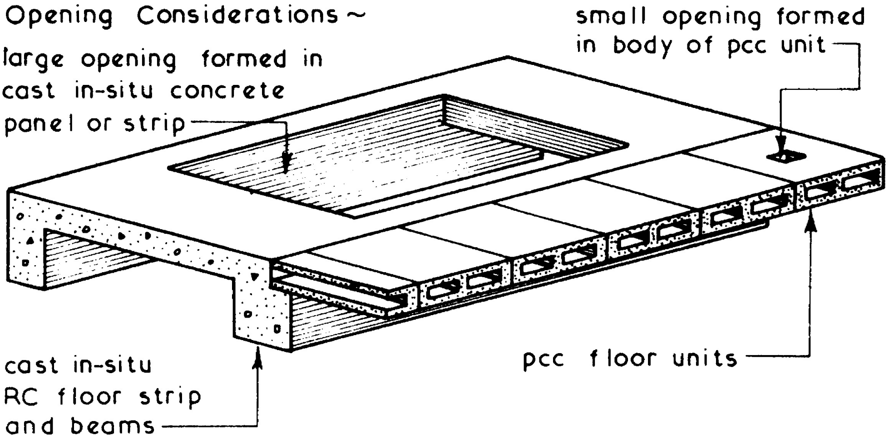
STEEL SUPPORT   CONCRETE SUPPORT   MASONRY SUPPORT

NB. spalling to end of pcc unit and/or edge of support will reduce effective bearing length.

Lateral Restraint and Structural Continuity Considerations ~



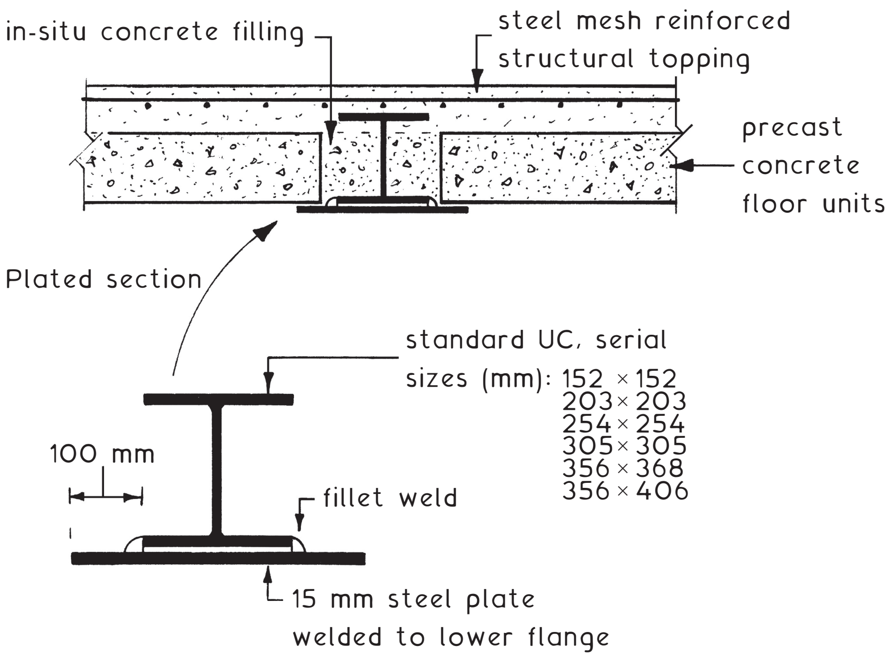
Opening Considerations ~



Steel fabricated beams can be used as an integral means of support for precast concrete floors. These are an overall depth and space-saving alternative compared to down-stand reinforced concrete beams or masonry walls. Only the lower steel flange of the steel beam is exposed.

To attain sufficient strength, a supplementary steel plate is welded to the bottom flange of standard UC sections. This produces a type of compound or plated section that is supported by the main structural frame.

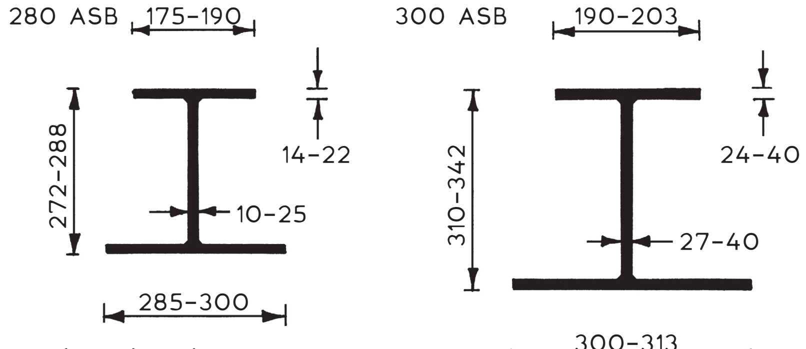
Floor support



Note: For fillet weld details and other welding applications, see page 672.

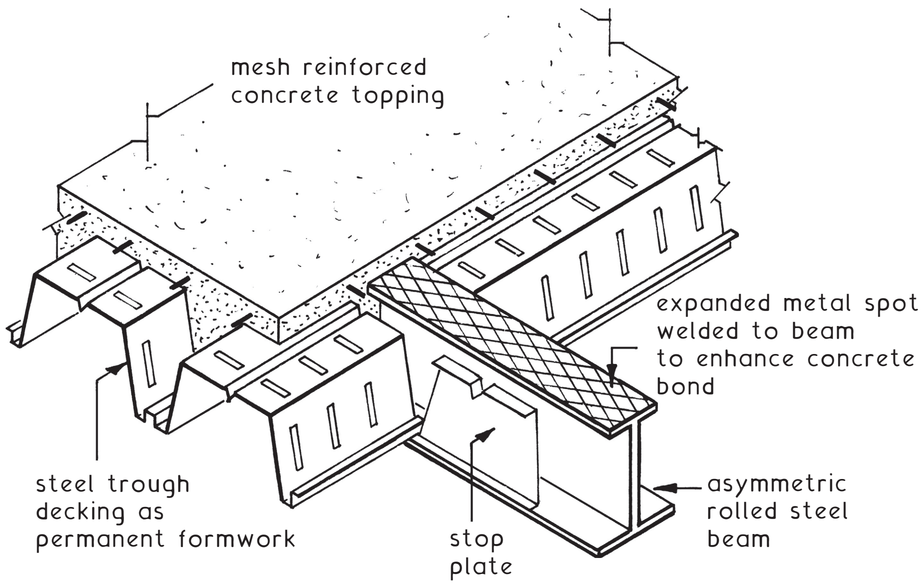
A standard manufactured steel beam with similar applications to the plated UC shown on the previous page. This purpose-made alternative is used with lightweight flooring units, such as precast concrete hollow core slabs and metal section decking of the type shown on page 621.

Standard production serial size range ~



Dimensions in mm

Application ~

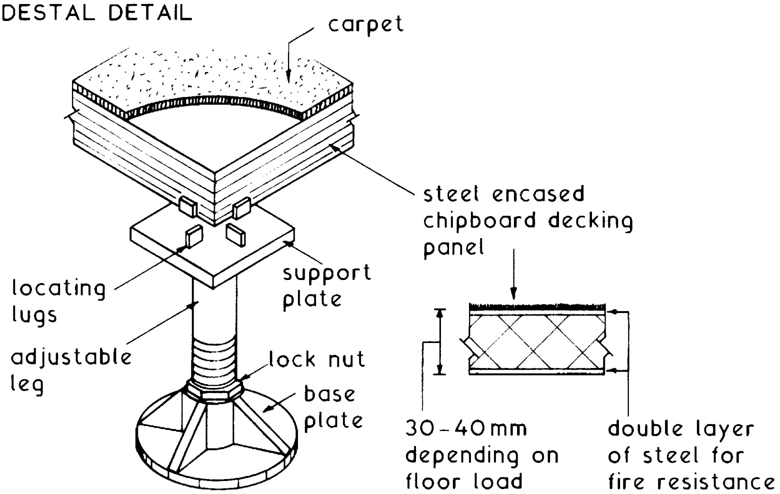


Raised Flooring ~ developed in response to the high-tech boom of the 1970s. It has proved expedient in accommodating computer and communications cabling as well as numerous other established services. The system is a combination of adjustable floor pedestals, supporting a variety of decking materials. Pedestal height ranges from as little as 30 mm up to about 600 mm, although greater heights are possible at the expense of structural floor levels. Decking is usually in loose fit squares of 600 mm, but may be sheet plywood or particleboard screwed direct to closer spaced pedestal support plates on to joists bearing on pedestals.

Cavity fire stops are required between decking and structural floor at appropriate intervals (see Building Regulations, A D B, Volume 2, Section 9).

Application ~

PEDESTAL DETAIL



FLOOR SECTION